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Customer-centric circular economy as-a-service decision-making: Machine learning-driven open innovation in food service

Tutur Wicaksono [a], Marhadi Marhadi [b,c,*], Agustinus Fritz Wijaya [d], Velly Anatasia [e], Krisztina Taralik [f]

[a] School of Business and Management, Universitas Ciputra, Indonesia
[b] Doctoral School of Economic and Regional Sciences, Hungarian University of Agriculture and Life Sciences, Hungary
[c] Department of Management, Faculty of Business and Economics, Universitas Riau, Indonesia
[d] Department of Informatics, Faculty of Technology and Design, Universitas Bunda Mulia, Jakarta, Indonesia
[e] Department of Management, Faculty of Social Sciences and Humanities, Universitas Bunda Mulia, Jakarta, Indonesia
[f] Department of Agricultural logistic, Commerce, Marketing, Hungarian University of Agriculture and Life Sciences, Hungary

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ABSTRACT

This study addresses a key research gap by developing a machine learning-driven open innovation framework for prioritizing Circular Economy as a Service (CEaaS) measures in the food service sector. Conventional sustainability strategies often fail to integrate dynamic customer preferences with scalable circular economy practices. To bridge this gap, this research employs Random Forest classification and feature selection to assess the impact of sixteen CEaaS measures on customer satisfaction across Indonesian and Hungarian food service markets. The framework is grounded in PRISMA-guided literature review, customer surveys, and machine learning-based decision analytics. Sustainable sourcing, reusable packaging systems, and circular loyalty programs emerge as the most critical CEaaS measures, reflecting global consumer priorities for ethical procurement, waste minimization, and behavior-based engagement. Notably, surplus food redistribution and food waste upcycling services rank consistently high, underscoring growing expectations for visible circularity and social value. The cross-country model validation confirms the framework's robustness and generalizability. By integrating circular economy principles, customer insights, and machine learning, the study advances open innovation theory and provides actionable guidance for sustainable transformation in the food service sector. The findings support Sustainable Development Goals related to responsible consumption, climate action, industry innovation, and global partnerships.

1. Introduction

The global food service industry plays a vital role in modern economies, driving employment, economic growth, and cultural exchange (Recuero-Virto and Arróspide, 2024). However, this sector faces a critical contradiction, as it is a leading contributor to environmental degradation, generating escalating waste, greenhouse gas emissions, and resource depletion (Zhang et al., 2021). As the urgency to address environmental sustainability intensifies, food service businesses face mounting pressure to adopt circular economy (CE) principles. These principles focus on creating closed-loop systems by reducing waste, reusing resources, and recycling (Kurniawan et al., 2024). Despite growing awareness, the transition to circularity remains fraught with challenges, as businesses struggle to align operational sustainability with customer expectations, leading to a gap between their commitment to circularity and its actual implementation (Dorrego-Viera et al., 2025). Addressing these challenges is essential not only for the industry but also for global sustainability. In line with the United Nations Sustainable Development Goals, particularly SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 17 (Partnerships for the Goals), this research aims to develop strategies that support both environmental responsibility and long-term economic growth (Azmat et al., 2023).

The concept of Circular Economy as a Service (CEaaS) has emerged

* Corresponding author. Doctoral School of Economic and Regional Sciences, Hungarian University of Agriculture and Life Sciences, Hungary.
E-mail addresses: tutur.wicaksono@ciputra.ac.id (T. Wicaksono), marhadi.marhadi-@phd.uni-mate.hu, marhadi@lecturer.unri.ac.id (M. Marhadi), agustinus.wijaya@bundamulia.ac.id (A.F. Wijaya), vanatasia@bundamulia.ac.id (V. Anatasia), taralik.krisztina@uni-mate.hu (K. Taralik).

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as a promising solution, enabling businesses to adopt circular practices without significant upfront investments (Greer et al., 2020). The CEaaS framework offers access to circular solutions such as waste reduction technologies, reusable packaging systems, and resource recovery services, which are made available through subscription or pay-per-use models (Da Silva et al., 2024). However, the successful implementation of CEaaS in food services requires a deep understanding of customer needs and preferences, as well as the ability to prioritize measures that deliver both environmental and economic value (Rakesh and Mahendran, 2023). This challenge underscores the importance of open innovation, a collaborative approach that leverages external insights, particularly from customers, to drive innovation and sustainability (Yun et al., 2023, 2025).

Integrating customer insights into the design and prioritization of CEaaS measures is essential for ensuring market relevance and adoption (Ekman et al., 2020). For example, while reusable packaging systems may appeal to environmentally conscious consumers, they may face resistance from those who prioritize convenience or cost-effectiveness (Rabiu and Jaeger-Erben, 2023). This reflects the complexity of decision-making in CEaaS implementation, which requires balancing multiple criteria such as sustainability, customer satisfaction, and profitability (Adelodun et al., 2021). Traditional methods, such as multi-criteria decision analysis (MCDA), often struggle to address the dynamic and multifaceted nature of customer preferences, especially when working with large and complex datasets (Beinabadi et al., 2024).

The advancement of big data and machine learning techniques offers new opportunities to overcome these challenges. Algorithms like Random Forest (RF) are particularly suited for analyzing large datasets, uncovering patterns and relationships that may be overlooked by conventional methods (Mahboubi et al., 2024; Pritee et al., 2024). RF supports both feature selection and classification, allowing researchers to identify and prioritize the most impactful CEaaS measures based on their importance scores (Rabiee et al., 2024). By applying RF, businesses can streamline CEaaS development by focusing on measures aligned with customer priorities while enhancing sustainability outcomes (Temizceri and Kara, 2024).

Although there is growing interest in circular economy and machine learning, few studies have examined how these concepts intersect within the CEaaS context in food service. Existing research tends to focus on specific areas such as waste reduction, resource optimization, or supply chain efficiency, rather than a service-based circular economy framework (Damiani et al., 2021). Moreover, conventional approaches remain dominant and often overlook the potential of machine learning to support scalable and adaptive circular economy solutions.

Recent developments in predictive analytics, IoT-enabled smart systems, and AI-based decision tools provide promising means to advance circularity in food services (Lakhouit, 2025). However, limited research has explored how machine learning can be integrated into service-oriented frameworks to enable closed-loop systems, reduce food waste, and improve sustainability. Similarly, the role of open innovation in co-developing CEaaS models with customers remains insufficiently studied, especially in the food service context (Fang et al., 2024; Dantas et al., 2022).

Indonesia and Hungary were selected to represent two distinct economic and regulatory contexts. Indonesia is an emerging economy in Southeast Asia with a large informal food service sector, while Hungary is a developed economy operating within the structured regulatory environment of the European Union. Both countries face similar challenges in circular economy adoption in food services, including food waste, resource inefficiency, and limited CE integration (Pribadi et al., 2025; Sanjuan-Delmás et al., 2021). However, their contrasting levels of economic development, digital readiness, and policy enforcement create a valuable setting for testing the robustness and relevance of CEaaS prioritization. This combination supports a comparative analysis and enhances the international applicability of the findings.

This study addresses these gaps by proposing a machine learning-driven open innovation framework for constructing CEaaS in food services. The research aims to answer the following questions:

RQ 1: How can machine learning models be integrated into an open innovation approach to develop a customer-centric Circular Economy as a Service (CEaaS) framework in food services?
RQ 2: What are the prioritized CEaaS measures that best align with customer satisfaction in food services?

To answer this question, the study pursues the following objectives:

1. Develop an integrated framework combining open innovation and machine learning for customer-centric CEaaS solutions.
2. Identify potential CEaaS measures for food service providers through a systematic literature review using the PRISMA framework.
3. Utilize Random Forest classifier to customer survey data to assess and validate the performance of the proposed framework.
4. Utilize Random Forest-based feature selection on customer survey data to rank the priority CEaaS measures according to their significance on customer satisfaction level.

By integrating circular economy principles, open innovation, and machine learning, this research provides a robust foundation for advancing sustainability in the food service industry. The proposed framework not only addresses environmental challenges but also enhances customer satisfaction and business competitiveness in a sustainability-driven market. The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on circular economy as a service in the food service industry, open innovation in circular economy, and machine learning for circular economy decision-making. Section 3 outlines the research methodology, detailing the use of the PRISMA Framework, Random Forest Classifier, and Random Forest feature selection. Section 4 presents the results and discussion, while Section 5 discusses the conclusions and their implications for theory and practice. Finally, Section 6 concludes by summarizing the contributions and suggesting directions for future research.

2. Literature review

2.1. Circular economy as a service: concept and application in the food service industry

The circular economy (CE) represents a transformative shift from the traditional linear “take-make-dispose” paradigm to a regenerative system that emphasizes resource efficiency, waste minimization, and systemic resilience (Geissdoerfer et al., 2016; Hussain et al., 2023). This fundamental change is especially significant for the food service industry, which contributes substantially to greenhouse gas emissions and food waste (Carino et al., 2020). By embracing the principles of reducing, reusing, recycling, and regenerating, CE provides a structured framework for addressing environmental challenges through closed-loop systems that optimize resource utilization (Acerbi and Taisch, 2020). Building on these principles, Circular Economy as a Service (CEaaS) extends traditional CE approaches by incorporating servitization. Unlike conventional business models that rely primarily on product sales, CEaaS focuses on performance-based services where revenue is directly tied to service outcomes (Patwa et al., 2020). This shift embeds long-term resource optimization and waste minimization into daily operations, enhancing both sustainability and economic efficiency (Kanda et al., 2024).

Numerous studies have demonstrated the effectiveness of CEaaS in reducing waste and improving operational efficiency in food-related industries. For example, reusable container systems have significantly cut costs by optimizing inventory management, reducing the frequency of supplier visits, and minimizing associated expenses (Becker et al., 2023). Similarly, integrating microalgae cultivation with anaerobic

digestion has transformed food waste into biogas, electricity, and biodiesel, while recycling digestate effluent for microalgae growth and fertilizer production establishes a holistic energy and waste management framework (Ro et al., 2024). Furthermore, CE strategies are essential for addressing the water-food-energy nexus by enhancing water recovery and reducing waste (Samberger, 2022). Varjani et al. (2023) further emphasize the use of contemporary technologies, such as metagenomics and AI-driven solutions, to optimize resource utilization, enhance food waste valorization, and mitigate environmental impacts from plastic pollution and traditional waste disposal.

Digital platforms also play a crucial role in enabling CEaaS in the food service industry. Platforms like Too Good To Go (TGTG) connect businesses with surplus food and consumers across more than 17 countries, simultaneously reducing food waste and alleviating food insecurity (Magno and Cassia, 2024; Too Good To Go, 2023). The success of these technology-driven solutions depends on effective collaboration among policymakers, businesses, and consumers, as well as supportive regulatory frameworks and active stakeholder engagement.

Regulatory frameworks facilitate the adoption of CE practices. Global initiatives, such as the European Union's Circular Economy Action Plan, have set ambitious waste reduction targets (European Commission, 2020; Moschen-Schimek et al., 2023), while Southeast Asia's Framework for Circular Economy for the ASEAN Economic Community (AEC) reflects a regional commitment to resource efficiency (ASEAN, 2021; Herrador and Van, 2023). However, fragmented policies and varying enforcement levels continue to hinder implementation. Challenges such as inconsistent date labeling, variable donation incentives, and uneven landfill mandates discourage stakeholder collaboration and call for harmonized policies and standardized incentives (Ryen and Babbitt, 2022).

Behavioral barriers also complicate the transition to CE in the food service industry. Consumers often exhibit a "single-use bias" by prioritizing convenience over sustainability, even as awareness of environmental consequences grows (Rabiu and Jaeger-Erben, 2023). In parallel, small and medium enterprises (SMEs) frequently face financial constraints that limit investments in Circular Economy technologies due to high upfront costs (Worakittikul et al., 2025). These challenges suggest the need for hybrid framework that combine regulatory incentives with behavioral interventions. For instance, tax breaks for zero-waste certifications and consumer education initiatives, such as carbon footprint labeling, can encourage sustainable purchasing decisions (Huang et al., 2024). Regulatory measures that limit disposable materials can also shift consumption patterns (Borg et al., 2022). For example, replacing ten percent of disposable materials with reusable alternatives may prevent most waste from causing environmental damage, particularly in marine ecosystems (WEF, 2021). These points underscore the importance of targeted policy frameworks in driving both consumer and business engagement in circular economy practices.

Finally, scaling CE in the food service industry requires redefining value chains through sustainable sourcing, transparent supply chains, and technological innovations (Al-Obadi et al., 2022). Farm-to-table networks that prioritize local sourcing help reduce transportation emissions and minimize food spoilage, thereby reinforcing CE principles at multiple stages of the supply chain (Camilleri, 2024). In addition, blockchain-enabled traceability systems enhance transparency and accountability in sustainable sourcing while reducing inefficiencies in circular supply chains (Mothafar et al., 2024). Despite operational challenges in an industry marked by low-profit margins and cost sensitivity (Qahtan et al., 2024), the transformative potential of CE in food services is significant if businesses, policymakers, and consumers collectively embrace sustainability.

2.2. Open innovation in circular economy: co-creation and customer-centric solution

Open innovation has emerged as a transformative paradigm that bridges the gap between environmental imperatives and market realities in the food service industry (Leitão et al., 2020). Grounded in the theory that firms must leverage external knowledge and partnerships to accelerate innovation (Chesbrough, 2019; Yun et al., 2024), open innovation offers a critical framework for addressing the systemic complexity of circular economy transitions (Srisathan et al., 2023). Unlike traditional closed innovation approaches that rely solely on internal research and development and often yield misaligned sustainability solutions, open innovation fosters co-creation by integrating diverse stakeholders, including customers, into the innovation process (Tsai et al., 2022). This approach aligns with the multi-level perspective of socio-technical transitions, which considers the interplay between niche innovations (such as CEaaS), established industry norms, and broader sustainability pressures (Liu and Stephens, 2019). By involving external stakeholders in the design of CEaaS, businesses can better navigate these interdependencies and align technical feasibility with consumer expectations (Johnson, 2022).

The theoretical underpinnings of open innovation in circular economy are further enriched by the resource-based view and stakeholder theory. The resource-based view argues that competitive advantage stems from unique, valuable resources, which in circular economy as a service contexts include external expertise such as waste management innovators and customer insights (Csedő et al., 2023). Stakeholder theory asserts that firms must address the needs of all stakeholders, including customers, regulators, and communities, to achieve long-term sustainability (Durmaz et al., 2021). Together, these theories justify the need for open innovation in circular economy as a service, as external collaboration becomes a strategic resource that balances competing stakeholder interests to deliver solutions that are both environmentally effective and market relevant (Eisenreich et al., 2022).

A practical example of open innovation in food services is the collaboration between restaurants and local farming communities to establish a zero-waste supply chain. In this approach, restaurants redesign procurement processes by repurposing surplus ingredients, replacing single-use packaging, and co-developing sustainable menu options based on customer feedback. Tasting events and workshops further ensure that these offerings align with consumer preferences while maintaining convenience and culinary appeal (Camilleri, 2024). This integration of circular economy principles into both operations and customer engagement demonstrates value co-creation through iterative producer-consumer collaboration (John and Supramaniam, 2023).

At the core of CEaaS driven by open innovation is customer-centricity, which redefines end-users as co-designers of circular solutions (Gajdzik et al., 2024). This perspective aligns with design thinking methodologies that emphasize empathy, ideation, and prototyping in addressing complex sustainability challenges (Imjai et al., 2024). For example, the divergence between environmentally conscious users who favor reusable packaging and time-constrained segments that prefer single-use options highlights the need for open innovation to reconcile conflicting priorities (Sandhu et al., 2020). By integrating customer insights, businesses can develop segmented CEaaS offerings tailored to diverse needs, thereby increasing adoption rates (Bocken and Konietzko, 2022).

Despite its promise, implementing open innovation in CEaaS faces significant challenges. Transaction cost theory explains that firms may resist external collaboration due to perceived risks such as intellectual property infringement and coordination costs (Audretsch and Belitski, 2022). Moreover, behavioral inertia in both organizational routines and consumer habits often impedes the adoption of circular practices. Kanwal et al. (2024) note that ingrained consumption and disposal behaviors, reinforced by cultural norms surrounding food waste, hinder sustainable waste management despite the potential benefits of upcycling and advanced reduction technologies. Similarly, Möslinger et al. (2023) observe that food service systems rarely adopt industrial symbiosis and wastewater reuse, with future strategies likely to focus more on circular business models, sustainable materials, and waste

prevention.

2.3. Machine learning for circular economy decision-making

Machine learning is revolutionizing decision-making in the circular economy, particularly in the food service sector (Kanwal et al., 2024). By leveraging advanced data analytics, Machine learning transcends traditional linear fameworks and offers an adaptive framework to balance environmental impact, economic feasibility, and consumer preferences (López-Guajardo et al., 2021). This capability is essential in industries facing resource constraints and rapidly changing market conditions, where both operational flexibility and long-term sustainability are crucial (Behera et al., 2023).

The theoretical basis for applying machine learning in CE decision-making is grounded in complexity theory and systems thinking. Complexity theory posits that sustainability challenges, such as waste reduction and resource optimization, are inherently complex and nonlinear, necessitating adaptive and iterative solutions (Tetteh et al., 2024). Among various machine learning algorithms, Random Forest is particularly well suited for processing large multidimensional datasets, identifying intricate nonlinear relationships, and performing effective feature selection (Murthy et al., 2024; Xuan et al., 2021). These capabilities enable businesses to prioritize sustainability measures more accurately, as demonstrated in predictive food waste analytics (Tsui et al., 2022) and inventory optimization (Kanwal et al., 2024).

Decision theory further supports the application of machine learning by providing a structured framework for evaluating multiple alternatives under uncertainty (Albahri et al., 2023). Ensemble methods, such as those employed in Random Forest, aggregate multiple decision trees to reduce variance and mitigate overfitting, resulting in more reliable predictor importance scores (Devasahayam and Albijanic, 2023). This methodological rigor is especially valuable for integrating machine learning into CEaaS framework, where it enhances operational strategies and aligns business practices with evolving market demands and regulatory requirements (Danish and Senjyu, 2023).

Furthermore, systems theory emphasizes the interdependence of diverse data sources, including customer surveys, operational records, and market trends, in the practical application of machine learning in CEaaS. By integrating these datasets, machine learning models can develop adaptive decision-making frameworks that enable real-time monitoring and adjustments. For example, IoT-enabled systems combined with machine learning algorithms allow for real-time tracking of resource consumption, optimization of energy and water usage, and waste minimization (Javaid et al., 2022). In addition, predictive analytics models that forecast food waste patterns can further inform inventory management and support waste reduction efforts (Zatsu et al., 2024).

3. Methodology

This research proposes a framework to identify and prioritize Circular Economy as a Service (CEaaS) measures in the food service industry, with customer satisfaction level as the target variable. The methodology begins with data collection through a systematic literature review conducted under the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to ensure a rigorous and transparent selection process (Ly et al., 2024). Literature was sourced from the Scopus and Web of Science databases using the search query ALL (“Circular economy” “food service”). Articles addressing general sustainability practices were excluded, retaining only those focused specifically on CEaaS measures within the food service sector. During the screening phase, conference papers, book chapters, and non-academic publications were discarded due to quality concerns. After reviewing abstracts, introductions, results, and conclusions, 130 articles were excluded, leaving 14 core articles that formed the foundation for identifying potential CEaaS measures (see Fig. 1).

Following the literature review, a customer survey was conducted to collect data on the perceived importance of each CEaaS measure and its impact on customer satisfaction. The survey consisted of three sections. The first section collected demographic and behavioral information, including age, gender, experience, education level, occupation, and purchasing frequency. The second section asked respondents to rate the importance of each CEaaS measure on a five-point Likert scale ranging from 1 (Not Important at All) to 5 (Extremely Important) in relation to their satisfaction during the purchasing process. The third section asked respondents to evaluate how the implementation of each measure would

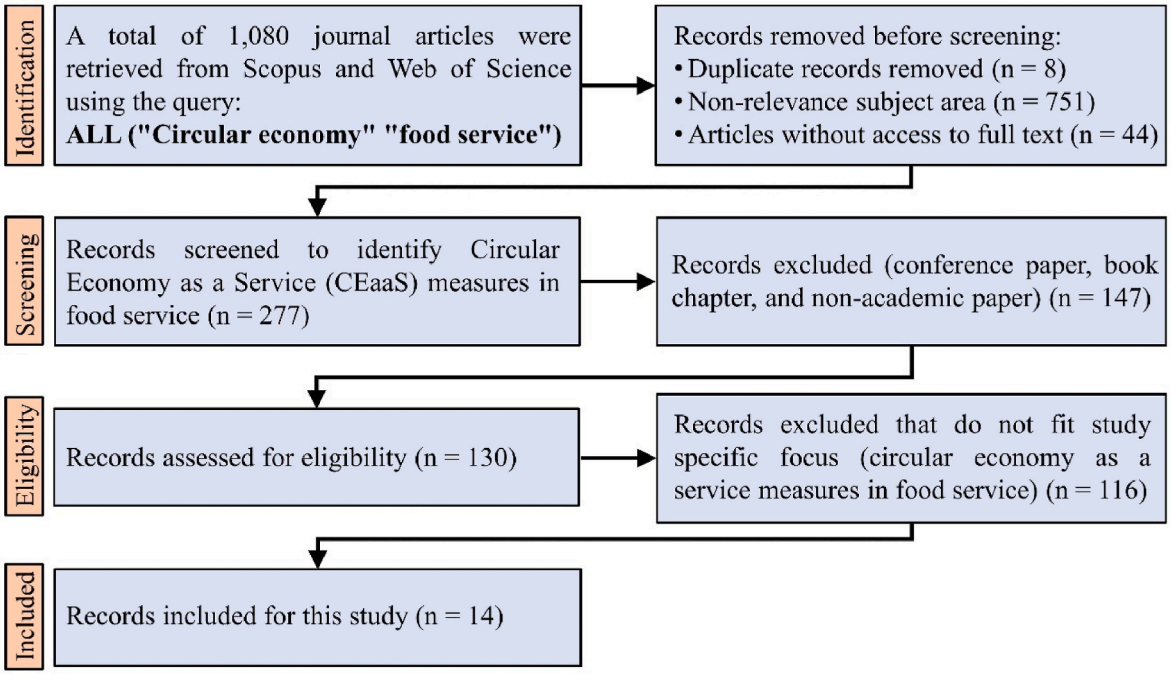


Fig. 1. Sistematic literature review through PRISMA framework.

influence their overall satisfaction, using the same scale.

The food service industry in Indonesia is largely dual-structured, with a dominant informal sector composed of street vendors and small-scale eateries, alongside a rapidly growing formal sector driven by urbanization and digital platforms. Regulatory enforcement and sustainability integration are limited, particularly outside major urban centers. In contrast, Hungary's food service sector is predominantly formal, characterized by chain restaurants and institutional providers, and governed by strict European Union regulations related to sustainability and waste management. These structural differences offer a meaningful comparative context for analyzing CEaaS adoption and prioritization across diverse regulatory and socio-economic environments.

A total of 1428 responses were collected, consisting of 820 from Indonesia and 608 from Hungary. The survey was conducted using a hybrid approach. Most respondents completed a self-administered online questionnaire via Google Forms, accessed through a QR code or barcode, while a smaller portion were invited via email and social media groups. Target participants were individuals who had recently used food service establishments such as restaurants, food stalls, or cafés. A purposive sampling method was employed, with inclusion criteria requiring participants to have made a food service purchase within the past 30 days. Participation was voluntary and anonymous, and informed consent was obtained prior to the start of the survey. Rather than enabling a direct comparison, the cross-country approach enhances the generalizability of findings and supports industry-wide recommendations for sustainable practices.

Given the dataset's size, machine learning algorithms were selected for their efficiency and accuracy in handling large-scale data, compared to traditional methods. The second stage involved data preparation, including cleaning and splitting. The dataset was processed to remove missing values and saved in CSV format. Independent variables (CEaaS measures) were separated from the dependent variable (customer satisfaction level). The methodology further employs two machine learning techniques: the Random Forest classifier and Random Forest-based feature selection. The Random Forest classifier aggregates predictions from multiple decision trees through majority voting (Sun et al., 2023). The process is outlined in the following steps:

Step 1: Boostraping and data sampling

The step begins by generating n_{tree} bootstrap samples from the original dataset. Let $X = \{x_1, x_2, \ldots, x_d\}$ represent the set of independent variables (CEaaS measures), and $Y = \{y_1, y_2, \ldots, y_d\}$ represent the corresponding target variable (set of customer satisfaction level responses). In bootstrapping, each new sample (X_c, Y_c) is created by random sampling with replacement from (X, Y), resulting in new datasets for each bootstrap sample. This is mathematically represented as:

$$(X_c, Y_c) = Bootstrap\ (X, Y)\ for\ c = 1, 2, \ldots, n \tag{1}$$

Each bootstrap sample (X_c, Y_c) is used to train a decision tree in the Random Forest. The bootstrapping process ensures that each tree is trained on a slightly different subset of the data, which helps to reduce overfitting and improve the model's **reliability.**

Step 2: Decision tree construction

For each bootstrap sample (X_c, Y_c), a decision tree is trained. Each tree is built by selecting a random subset of features at each split, which promotes diversity among the trees.

Step 3: Prediction aggregation

For a new observation, the prediction is made by aggregating the results from all n_{tree} decision trees using majority voting. The final prediction is the mode of all individual tree predictions:

$$\widehat{y} = mode(T_1(x), T_2(x), \ldots, T_{n_{tree}}(x)) \tag{2}$$

Step 4: Out-of-Bag (OOB) Error Calculation

The OOB error is calculated to evaluate the model performance without using a separate validation set. Each data point is tested on the decision trees for which it was not included in the bootstrap sample, and the OOB error is computed as:

$$OOB_{error} = \frac{1}{n_{OOB}} \sum_{i=1}^{n_{OOB}} \mathbb{I}(y_i \neq \widehat{y}_i) \tag{3}$$

where is $\mathbb{I}$ the indicator function that returns 1 if the observation is misclassified.

Next, Random Forest feature selection evaluates the importance of each CEaaS measure based on its impact on model error. Importance scores are determined by measuring the increase in error when a feature is permuted (Dunne et al., 2023). Measures causing the highest error increase are ranked as the most influential and prioritized accordingly. The steps are as follows:

Step 1: Tree construction

A set of K decision trees is constructed using the original dataset (X, Y)

Step 2: Calculate OOB error

For each tree i in the forest, calculate the out-of-bag (OOB) error using the unmodified OOB dataset, denoted as $ErrorOOB^{(i)}$. This provides a baseline measure of error for each decision tree in the forest.

Step 3: Permutation importance

Each feature's importance is calculated by randomly permuting its values in the OOB dataset and computing the error difference. The importance score for each feature I_{X_j} is defined as:

$$I_{X_j} = \frac{1}{n_{tree}} \sum_{i=1}^{n_{tree}} \left(ErrorOOB_j^{(i)} - ErrorOOB^{(i)} \right) \tag{4}$$

The performance of the Random Forest classifier model is evaluated using accuracy and F-score metrics to ensure reliability and validity. Accuracy measures the proportion of correct predictions, while the F-score balances precision and recall, assessing the model's effectiveness in capturing relevant patterns related to customer satisfaction. Performance evaluation is conducted before and after feature selection to assess improvements. The metrics are calculated using the following equations:

$$Accuracy = \frac{T_p + T_n}{T_p + F_p + F_n + T_n} \tag{5}$$

$$Fscore = \frac{Precision \times Recall}{Precision + Recall} \tag{6}$$

For each country-specific sub-sample, the Random Forest model was trained using an 80/20 train-test split, accompanied by 5-fold cross-validation to ensure reliability and reduce overfitting (Adewale et al., 2024). For the pooled dataset, robustness was assessed through cross-country validation using two scenarios: in Scenario 1, the model was trained on the Indonesian sub-sample and tested on the Hungarian sub-sample; in Scenario 2, the model was trained on the Hungarian sub-sample and tested on the Indonesian sub-sample. All modeling procedures were implemented in Python 3.13 via Google Colab, using

the Scikit-learn library. The computational setup included a 2.20 GHz processor, 16 GB RAM, a 4 GB graphics card, and a 64-bit operating system. Hyperparameter tuning determined that 700 decision trees offered an optimal balance between predictive accuracy and computational efficiency (Sun et al., 2023). The Gini index was selected as the splitting criterion, following best practices in classification-based feature importance modeling (Majidian et al., 2024). The Random Forest algorithm was selected for its robustness against overfitting and its suitability for high-dimensional, decision-oriented problems (Yang et al., 2024). The full research framework, including data collection, model development, and prioritization, is presented in Fig. 2.

4. Results and discussion

4.1. Profile respondents

The sample of 1428 respondents comprised food service customers from Indonesia (820 respondents, 57.42 %) and Hungary (608 respondents, 42.58 %), providing perspectives from both an emerging and a developed market. Gender distribution was well-balanced across both countries. In Indonesia, 373 respondents (45.5 %) were male and 447 (54.5 %) were female, while in Hungary, 295 respondents (48.5 %) were male and 313 (51.5 %) were female.

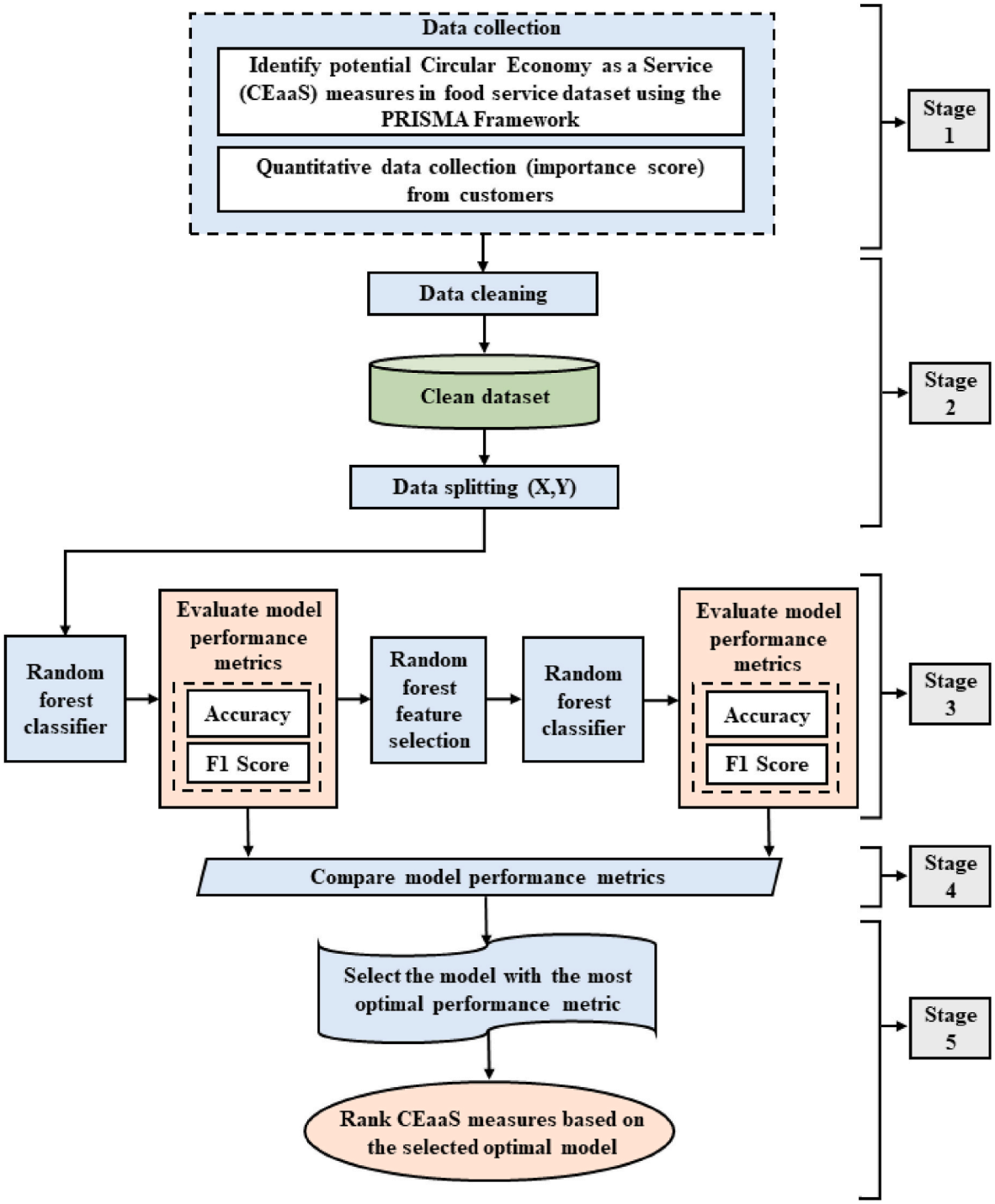


Fig. 2. Research framework.

Age distribution followed similar patterns in both countries, with some variations. In Indonesia, the largest age group was 30–44 years (339 respondents, 41.3 %), followed by 18–29 years (266 respondents, 32.4 %). Respondents aged 45–59 accounted for 17.6 % (144 respondents), while those aged 60 or older represented the smallest group (71 respondents, 8.7 %). Hungary showed a comparable distribution, with the 30–44 age group being the largest (285 respondents, 46.9 %), followed by 18–29 years (152 respondents, 25.0 %). The 45–59 and 60+ age groups constituted 18.6 % (113 respondents) and 9.5 % (58 respondents), respectively.

Educational backgrounds differed slightly between the two countries. In Indonesia, the largest proportion of respondents held a Bachelor's degree (348 respondents, 42.4 %), followed by high school graduates (210 respondents, 25.6 %). Associate degrees were held by 152 respondents (18.5 %), while Master's and Doctorate degrees accounted for 10.5 % (86 respondents) and 2.9 % (24 respondents), respectively. In Hungary, Bachelor's degrees were also the most common (246 respondents, 40.5 %), followed by high school diplomas (172 respondents, 28.3 %). Associate degrees were held by 114 respondents (18.8 %), while Master's degrees represented 11.7 % (71 respondents). Only 0.8 % (5 respondents) held a Doctorate.

Employment status revealed that the majority of respondents in both countries were employed, with 480 individuals (58.5 %) in Indonesia and 396 individuals (65.1 %) in Hungary. Students constituted 16.5 % (135 respondents) in Indonesia and 17.4 % (106 respondents) in Hungary. Self-employed individuals represented 12.9 % (106 respondents) in Indonesia and 12.7 % (77 respondents) in Hungary. Unemployment rates differed more noticeably, with 10.6 % (87 respondents) in Indonesia compared to only 3.8 % (23 respondents) in Hungary. Retirees were the smallest group in both countries, comprising 1.5 % in Indonesia and 1.0 % in Hungary.

Food service purchase frequency followed similar trends in both markets. The most common category was "Sometimes (2–3 times a month)," reported by 250 respondents (30.5 %) in Indonesia and 189 (31.1 %) in Hungary. "Often (once a week)" was the second most frequent, with 239 respondents (29.1 %) in Indonesia and 178 (29.3 %) in Hungary. "Very often (multiple times a week)" was indicated by 199 respondents (24.3 %) in Indonesia and 135 (22.2 %) in Hungary, while "Rarely (once a month or less)" was reported by 132 respondents (16.1 %) in Indonesia and 106 (17.4 %) in Hungary. These findings are summarized in Table 1, which provides a comprehensive breakdown of demographic characteristics and purchasing behaviors across the two countries.

Table 1
Survey respondent demographic.

Demographic variables		Indonesia (n = 820)	Hungary (n = 608)
Gender	Male	373 (45.5 %)	295 (48.5 %)
	Female	447 (54.5 %)	313 (51.5 %)
Age	18–29	266 (32.4 %)	152 (25.0 %)
	30–44	339 (41.3 %)	285 (46.9 %)
	45–59	144 (17.6 %)	113 (18.6 %)
	60 or older	71 (8.7 %)	58 (9.5 %)
Education	High school	210 (25.6 %)	172 (28.3 %)
	Associate degree	152 (18.5 %)	114 (18.8 %)
	Bachelor's degree	348 (42.4 %)	246 (40.5 %)
	Master's degree	86 (10.5 %)	71 (11.7 %)
	Doctorate	24 (2.9 %)	5 (0.8 %)
Occupation	Student	135 (16.5 %)	106 (17.4 %)
	Employed	480 (58.5 %)	396 (65.1 %)
	Self-Employed	106 (12.9 %)	77 (12.7 %)
	Retired	12 (1.5 %)	6 (1 %)
	Unemployed	87 (10.6 %)	23 (3.8 %)
Customer food service purchase frequency	Rarely (once a month or less)	132 (16.1 %)	106 (17.4 %)
	Sometimes (2–3 times a month)	250 (30.5 %)	189 (31.1 %)
	Often (once a week)	239 (29.1 %)	178 (29.3 %)
	Very often (multiple times a week)	199 (24.3 %)	135 (22.2 %)

4.2. Measures identification

As mentioned earlier in methodology, Sixteen potential CEaaS measures affecting customer satisfaction in food service were identified through systematic literature review using PRISMA framework and were listed in Table 2. And were used in survey to collect the datasets in Indonesia and Hungary.

4.3. Model performance metrics and CEaaS measure prioritization

Three models were developed based on different datasets: the Indonesian sub-sample, the Hungarian sub-sample, and a pooled dataset combining both. Model performance was assessed using the Random Forest (RF) classifier, both before and after applying RF-based feature selection. In the initial phase, each sub-sample dataset was used to train its respective model using an 80/20 train-test split. The Indonesian model achieved an accuracy of 81.77 % (0.8177) and an F-score of 78.63 % (0.7863), while the Hungarian model recorded an accuracy of 82.44 % (0.8244) and an F-score of 79.27 % (0.7927). Further evaluation using 5-fold cross-validation yielded average performance metrics of 80.46 % accuracy and 77.38 % F-score for the Indonesian dataset, and 79.66 % accuracy and 78.51 % F-score for the Hungarian dataset. For the pooled dataset, a cross-country validation approach was implemented to assess generalizability. In Scenario 1, where the model was trained on the Indonesian dataset and tested on the Hungarian dataset, the model achieved an accuracy of 79.52 % (0.7952) and an F-score of 76.27 % (0.7627). In Scenario 2, trained on the Hungarian dataset and tested on the Indonesian dataset, the accuracy was 76.83 % (0.7683) with an F-score of 73.55 % (0.7355). These results indicate acceptable levels of model robustness both within individual country contexts and across cross-country scenarios.

Subsequently, Random Forest-based feature selection was applied separately to the Indonesian, Hungarian, and pooled datasets to determine the Relative Importance Score (RIS) of each of the 16 CEaaS measures. This process enabled ranking the measures based on their contribution to the classifier's predictive performance. The results are visualized in Fig. 3, which compares the prioritized CEaaS measures across the three datasets.

Following feature selection, each model was retrained using only the top seven CEaaS measures, which were identified as the most influential in achieving optimal model performance and relevance to the target variable. The hyperparameters used in the original training process were retained. This refinement led to a notable improvement in performance metrics.

In the Indonesian model, the top seven measures included: sustainable sourcing (CM6), reusable packaging systems (CM2), circular loyalty programs (CM16), surplus food redistribution (CM3), digital inventory management (CM7), food waste upcycling services (CM1), and carbon-neutral delivery services (CM8). Using the 80/20 train-test split, the F-score increased by 5.16 percentage points to 83.79 %, and accuracy improved by 3.82 points to 85.59 %. Under 5-fold cross-validation, the F-score increased by 4.82 points to 82.11 %, and accuracy improved by 5.47 points to 85.93 %. In the Hungarian model, the top seven measures were: circular loyalty programs (CM16), surplus food redistribution (CM3), sustainable sourcing (CM6), reusable packaging systems (CM2), food waste upcycling services (CM1), carbon-neutral delivery services (CM8), and digital inventory management (CM7). The 80/20 split showed an F-score increase of 4.92 points to 84.19 % and an accuracy improvement of 3.66 points to 86.10 %. With 5-fold cross-validation, the F-score increased by 4.73 points to 83.24 %, and accuracy rose by 5.57 points to 85.23 %. In the pooled model, the top seven measures included: sustainable sourcing (CM6), circular loyalty programs (CM16), reusable packaging systems (CM2), surplus food redistribution

Table 2
Potential of Circular Economy as a Service (CEaas) measures in food service.

CEaaS Measures (CM)	Description	References
Food waste upcycling services (CM1)	Transforms food waste into valuable products to provide customers with a sustainable and responsible food service experience.	Kanwal et al. (2024);
Reusable packaging systems (CM2)	This service offers reusable containers through a return-and-refill system to eliminate packaging waste and enhance customer convenience.	Camps-Posino et al. (2021)
Surplus food redistribution (CM3)	Redistributes surplus food to charities or consumers to provide customers with affordable options while reducing food waste.	Rut and Davies (2023)
Energy recovery from waste (CM4)	This service converts organic waste into renewable energy to ensure customers benefit from environmentally responsible food operations.	Iqbal et al. (2023);
Closed-loop water recycling systems (CM5)	This service treats and reuses water within food operations to support sustainability and responsible water management for customers.	Biswas et al. (2019)
Sustainable sourcing (CM6)	prioritizes local, regenerative, or surplus ingredients to provide customers with ethically sourced and environmentally responsible food options.	Kanwal et al. (2024);
Digital inventory management (CM7)	This service optimizes inventory through AI-driven systems to ensure customers receive fresher products and experience minimal food waste.	Omar et al. (2024);
Carbon-neutral delivery services (CM8)	This service offsets delivery emissions through sustainable practices to provide customers with environmentally responsible food delivery.	Hossain et al. (2023)
Sustainable food literacy programs (CM9)	This program offers workshops and digital tools to empower customers to make informed choices that support sustainable food consumption.	Teng and Chih (2022)
Modular equipment leasing (CM10)	Leased and upgradable kitchen equipment to ensure customers benefit from efficient and sustainable food preparation.	Da Silva et al. (2024)
Local food hubs (CM11)	Sources local, surplus, and imperfect produce to reduce waste, support local economies, cut emissions, and provide fresher ingredients through a low-carbon supply chain.	Son (2023)
Green cleaning (CM12)	This service uses environmentally friendly cleaning materials and methods to ensure customers benefit from a healthier, safer, and more sustainable food environment.	Jin et al. (2021)
End-of-life material take-back programs (CM13)	This program collects and recycles used materials from food operations to ensure customers contribute to a fully circular system.	Camps-Posino et al. (2021)
Regenerative agriculture partnerships (CM14)	This service connects businesses with regenerative farms to provide customers with products that actively support ecosystem restoration.	Cherubin et al. (2024)
Smart food waste tracking systems (CM15)	This service uses IoT technology to monitor food waste in real time to ensure customers benefit from sustainable resource management.	Singh et al. (2025)
Circular loyalty programs (CM16)	This program rewards customers for participating in sustainable practices to encourage responsible consumption and environmental engagement.	Krampe et al. (2025)

(CM3), food waste upcycling services (CM1), carbon-neutral delivery services (CM8), and digital inventory management (CM7). Under cross-country validation, Scenario 1 (trained on Indonesia, tested on Hungary) showed an F-score increase of 4.24 points to 80.51 % and accuracy improvement of 6.29 points to 85.81 %. Scenario 2 (trained on Hungary, tested on Indonesia) recorded an F-score increase of 6.79 points to 80.34 % and accuracy improvement of 6.44 points to 83.27 %. These findings underscore that the models achieved stronger robustness following feature selection. By eliminating nine less significant features, the models reduced noise in the datasets, thereby improving interpretability and predictive performance. The final selection of seven CEaaS measures in each model was justified by their high contribution to performance improvement, as indicated by increased F-score and accuracy. Detailed comparisons of model performance before and after feature selection are presented in Fig. 4.

The importance score of each CEaaS measure was derived using the feature importance outputs of the Random Forest algorithm. Each measure's significance level was calculated as a proportion of the total importance of the top seven measures. Table 3, Tables 4 and 5 present the detailed rankings, importance scores, and significance levels for the Indonesian, Hungarian, and pooled datasets, respectively.

The comparative analysis of CEaaS measure prioritizations across Indonesian, Hungarian, and pooled datasets reveals both convergence and divergence in sustainability preferences across national contexts, offering meaningful insights into region-specific consumer expectations as well as global trends in sustainable food service innovation. Notably, sustainable sourcing (CM6) emerges as the most critical measure in both the Indonesian and pooled datasets, with respective importance scores of 0.121333 (19.92 %) and 0.114727 (18.66 %). This consistent prioritization indicates a strong cross-national consensus regarding the importance of ethically sourced, traceable, and environmentally responsible food procurement practices. This finding echoes previous studies emphasizing consumer demand for supply chain transparency and ecological accountability (Hopkins et al., 2023; Wilkinson et al., 2024). In contrast, while CM6 still ranks highly in the Hungarian dataset (16.05 %), it is surpassed by circular loyalty programs (CM16) and surplus food redistribution (CM3), suggesting a more nuanced prioritization structure potentially shaped by distinct consumer values or regulatory pressures in the Hungarian context. The dominance of this measure indicates that consumers are increasingly demanding not only high-quality food products but also verifiable sustainability commitments from food service providers (Orden-Mejía and Moreno-Manzo, 2024; Liao et al., 2023).

Circular loyalty programs (CM16) are particularly notable for their top ranking in Hungary (17.34 %) and second-place position in the pooled dataset (17.21 %), indicating broad appeal across both national and integrated datasets. This measure also ranks third in Indonesia (14.83 %), suggesting that gamified, incentive-based mechanisms for encouraging sustainable behavior are viewed favorably across diverse consumer groups. The appeal of CM16 likely derives from its ability to bridge sustainability with consumer engagement, rewarding waste-reducing behaviors while strengthening brand-customer relationships (Krampe et al., 2025). These programs transform sustainability from a passive corporate value into an interactive, loyalty-enhancing experience.

Interestingly, reusable packaging systems (CM2) show substantial prioritization in Indonesia (16.84 %) and the pooled dataset (16.68 %), where they rank second and third, respectively. In Hungary, this measure ranks fourth (14.81 %), indicating relatively consistent perceived value across regions. The high priority attributed to CM2 highlights increasing public concern about single-use packaging waste and its environmental impacts. This preference aligns with ongoing consumer pressure for visible, product-level sustainability initiatives that minimize waste at the point of consumption (Bonfanti et al., 2025). Its ranking just below loyalty programs in Hungary may reflect a more behavior-centric interpretation of sustainability compared to the product-centric focus observed in the Indonesian context.

Measures related to food waste mitigation, such as surplus food

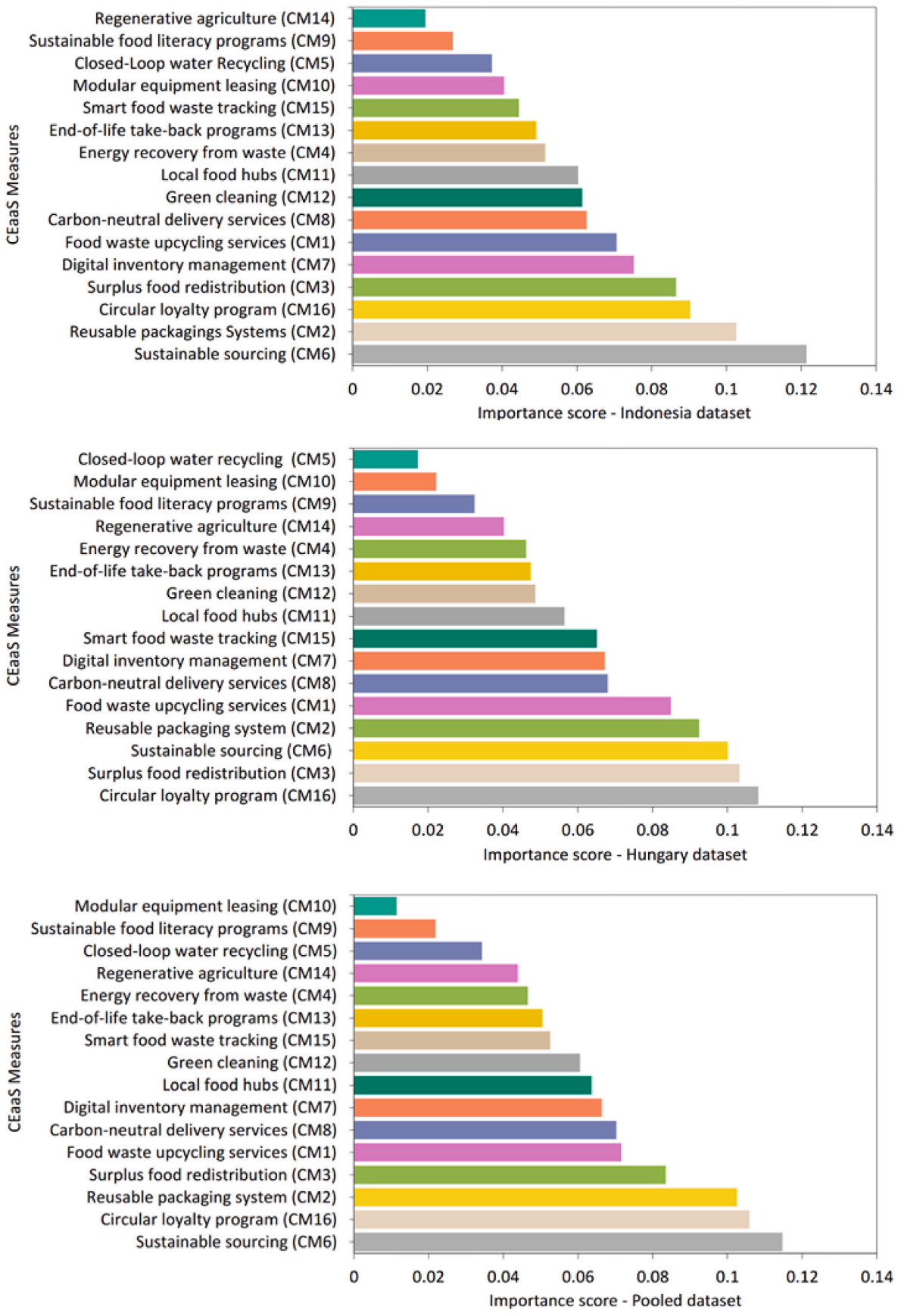


Fig. 3. CEaaS measures importance score.

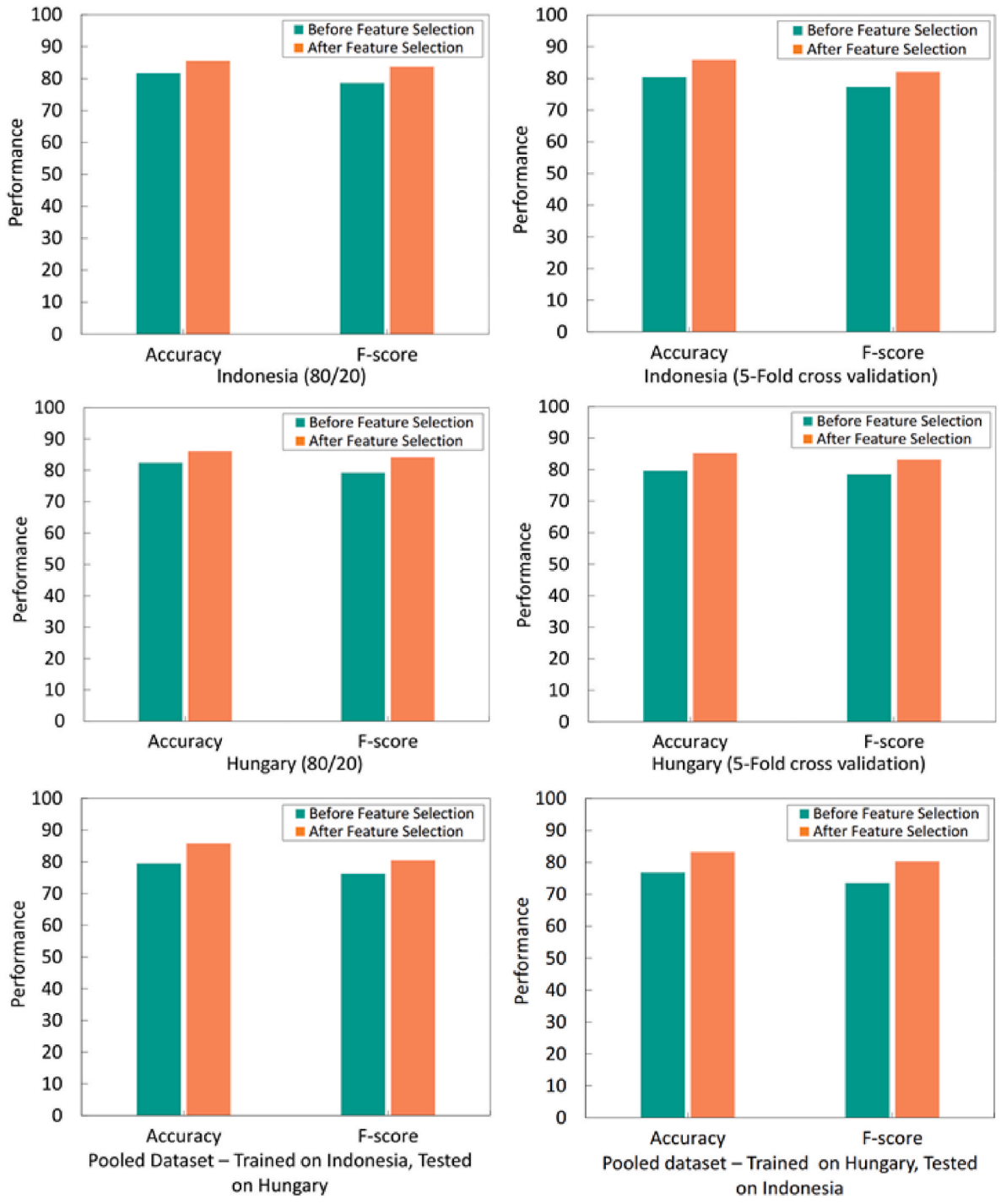


Fig. 4. Model performance metrics.

redistribution (CM3) and food waste upcycling services (CM1), consistently occupy mid-ranking positions across all datasets. CM3 ranks fourth in both Indonesia (14.20 %) and the pooled data (13.58 %), and second in Hungary (16.53 %). CM1, meanwhile, is ranked fifth in Hungary (13.60 %), sixth in Indonesia (11.59 %), and fifth in the pooled data (11.64 %). This recurring prominence underscores consumer awareness of and support for circular economy solutions that repurpose or redirect food waste. The prominence of CM3 in Hungary may also reflect cultural or institutional support for food-sharing systems or national food rescue regulations. In both cases, these measures reflect increasing expectations for businesses to move beyond traditional waste minimization and adopt proactive circular strategies (Kanwal et al., 2024).

The ranking of carbon-neutral delivery services (CM8) and digital inventory management (CM7) offers a particularly insightful contrast. While these measures are consistently placed at the bottom of the top-seven ranking across all datasets, their importance scores remain relatively high, particularly in Indonesia where CM8 ranks seventh (10.26 %) and CM7 ranks fifth (12.35 %). In Hungary and the pooled datasets, these measures switch positions but remain in the sixth and seventh spots, with similar significance levels (around 10–11 %). This pattern suggests that while these measures are perceived as essential enablers of sustainability, they are perhaps viewed more as backend or infrastructural strategies rather than consumer-facing priorities.

Table 3
CEaaS measures rank based on importance level (Indonesia dataset).

Priority CEaaS measures	Importance score	Significance level (%)	Ranks
Sustainable sourcing (CM6)	0.121333	0.199202	1
Reusable packaging system (CM2)	0.102564	0.168387	2
Circular loyalty program (CM16)	0.090326	0.148295	3
Surplus food redistribution (CM3)	0.086477	0.141976	4
Digital inventory management (CM7)	0.075225	0.123503	5
Food waste upcycling services (CM1)	0.070619	0.115941	6
Carbon-neutral delivery services (CM8)	0.062552	0.102696	7

Table 4
CEaaS measures rank based on importance level (Hungary dataset).

Priority CEaaS measures	Importance score	Significance level (%)	Ranks
Circular loyalty program (CM16)	0.108223	17.34	1
Surplus food redistribution (CM3)	0.103147	16.53	2
Sustainable sourcing (CM6)	0.100129	16.05	3
Reusable packaging system (CM2)	0.092433	14.81	4
Food waste upcycling services (CM1)	0.084877	13.60	5
Carbon-neutral delivery services (CM8)	0.068031	10.90	6
Digital inventory management (CM7)	0.067193	10.77	7

Table 5
CEaaS measures rank based on importance level (Pooled dataset).

Priority CEaaS measures	Importance score	Significance level (%)	Ranks
Sustainable sourcing (CM6)	0.114727	18.66	1
Circular loyalty program (CM16)	0.105816	17.21	2
Reusable packaging system (CM2)	0.102529	16.68	3
Surplus food redistribution (CM3)	0.083468	13.58	4
Food waste upcycling services (CM1)	0.071583	11.64	5
Carbon-neutral delivery services (CM8)	0.070273	11.43	6
Digital inventory management (CM7)	0.066358	10.79	7

Nevertheless, their inclusion in the top-tier measures across all datasets signals a growing consumer sensitivity to the environmental impacts of logistics and the role of intelligent systems in minimizing resource waste and ensuring service reliability (Milenković et al., 2024; Vhatkar et al., 2024).

From a broader perspective, the alignment across datasets reveals several globally resonant sustainability imperatives, particularly those related to ethical sourcing, packaging reuse, food waste recovery, and digitally enabled operational efficiency. However, the variation in top priorities across national datasets underscores the importance of context-sensitive strategies. Indonesian consumers appear more responsive to upstream environmental interventions such as sustainable sourcing and packaging, while Hungarian consumers place slightly greater emphasis on behaviorally engaging or socially focused measures such as loyalty programs and food redistribution.

These insights affirm that effective CEaaS strategies must incorporate both universal sustainability expectations and region-specific consumer preferences. A hybrid model that prioritizes traceable sourcing, visible waste-reduction systems, incentive-based behavioral design, and operational transparency will be best positioned to meet the multidimensional demands of today's food service consumers. Moreover, the high cumulative significance scores of the top seven measures in each dataset demonstrate the necessity for integrated, multi-pronged sustainability initiatives that span sourcing, engagement, logistics, and waste management. As food service businesses seek to enhance competitiveness and compliance in an era of climate-conscious consumption, aligning operations with these prioritized CEaaS measures will be instrumental in achieving sustainable transformation and long-term customer trust.

5. Conclusion

This study examined the prioritization of Circular Economy as a Service (CEaaS) measures in the food service sector using a Random Forest-based feature importance analysis across Indonesian, Hungarian, and pooled datasets. By comparing the top seven measures in each dataset, the findings reveal a clear convergence around certain sustainability imperatives, particularly sustainable sourcing, reusable packaging systems, circular loyalty programs, and food waste mitigation strategies. Despite national differences in ranking orders, these core measures consistently reflect global consumer expectations for ethically responsible, resource-efficient, and socially conscious food service practices.

Sustainable sourcing (CM6) emerged as the most critical measure in both the Indonesian and pooled datasets, while circular loyalty programs (CM16) were most emphasized in Hungary, highlighting both cross-national alignment and cultural variability in sustainability priorities. The consistently high ranking of food waste-related measures, such as surplus food redistribution (CM3) and food waste upcycling services (CM1), underscores a growing demand for visible circular practices that extend beyond environmental responsibility to include social value creation. Additionally, the inclusion of carbon-neutral delivery services (CM8) and digital inventory management (CM7) across all datasets reflects a broader recognition of the operational backbone required to support sustainable service delivery.

The results demonstrate that consumers increasingly value a comprehensive approach to sustainability, one that integrates environmental, social, and technological dimensions. Measures that enhance traceability, reduce packaging waste, incentivize responsible behavior, and minimize food loss are no longer viewed as supplementary efforts but as central to a competitive and future-ready food service strategy. While visible front-end sustainability initiatives remain highly valued, the growing importance of back-end operational efficiency signals a maturing consumer perspective on what constitutes meaningful sustainability.

By applying the Random Forest algorithm, this study provides an evidence-based understanding of feature prioritization that can inform strategic decision-making in food service innovation. The significance levels derived from model outputs offer practical guidance for stakeholders seeking to allocate resources and design interventions aligned with consumer values. The integration of machine learning enhances the objectivity and robustness of the prioritization process, reinforcing its utility for policymaking, corporate strategy, and service design in the context of circular economy transitions.

Overall, the findings reinforce the urgency for food service providers to adopt integrated CEaaS strategies that combine ethical sourcing, circular product systems, behavioral engagement tools, and digital operational infrastructures. Such strategies will be essential for achieving long-term customer satisfaction, regulatory compliance, and alignment with broader sustainability goals.

6. Research implications

6.1. Theoretical implication

This study contributes to the theoretical development of Circular Economy as a Service (CEaaS) in the food service domain by integrating machine learning with a structured prioritization framework rooted in customer-oriented sustainability. By applying the Random Forest algorithm for feature selection and prioritization, the research enhances methodological rigor in identifying CEaaS measures aligned with consumer preferences across diverse national contexts. It responds to the increasing demand for data-driven sustainability research that moves beyond conceptual generalizations toward empirically grounded, actionable insights (Boozary et al., 2025).

First, the study extends existing literature by systematically quantifying the relative importance of CEaaS measures through their contribution to model performance. Unlike prior works that treat sustainability practices as uniform initiatives, this study reveals regional heterogeneity in their perceived value. For example, the prominence of sustainable sourcing in both the Indonesian and pooled datasets, and the top ranking of circular loyalty programs in Hungary, reflect varied consumer expectations, emphasizing the importance of context-specific strategies. Second, the research refines our understanding of circular economy adoption by showing that consumer perceptions are influenced not only by front-facing interventions such as packaging or loyalty programs, but also by backend operations, including digital inventory systems and carbon-neutral logistics. This expands CEaaS's conceptual scope to include logistical and technological infrastructure as essential components of sustainable service delivery. Third, the study underscores the role of technological open innovation in advancing sustainability, highlighting how machine learning can catalyze systemic change in the food service sector (Rodrigues et al., 2023). The integration of feature selection with cross-country model validation demonstrates machine learning's potential not only for predictive performance but also for enabling generalizability across markets, offering a replicable framework for future research across sectors and regions.

6.2. Practical implication

The findings provide strategic guidance for food service businesses aiming to implement high-impact CEaaS initiatives. The consistent prioritization of sustainable sourcing across datasets underscores the need for ethical, traceable, and environmentally responsible procurement. Businesses are advised to collaborate with regenerative agriculture networks, local producers, and surplus food suppliers to enhance supply chain transparency and ecological stewardship (You et al., 2024). The strong performance of circular loyalty programs and reusable packaging systems highlights a dual imperative: incentivize consumer engagement through behavior-based rewards and reduce packaging-related waste. Food service firms should design loyalty schemes that promote low-waste behaviors and invest in scalable reusable packaging infrastructure that aligns with consumer convenience and sustainability preferences (Krampe et al., 2025; Bonfanti et al., 2025). The prominence of surplus food redistribution and food waste upcycling points to service innovation opportunities embedded in circular principles. Partnering with food banks, donation platforms, or upcycling startups allows businesses to cut environmental impact while generating social value and alternative revenue streams (Kanwal et al., 2024). The inclusion of carbon-neutral delivery and digital inventory management among the top-ranked measures indicates that operational innovations are essential to sustainability agendas. Businesses should pursue carbon mitigation strategies such as fleet electrification and carbon offset programs, while using AI-driven inventory tools to optimize freshness, reduce spoilage, and improve service reliability (Milenković et al., 2024; Vhatkar et al., 2024). These operational upgrades not only advance sustainability goals but also enhance brand credibility in a competitive, environmentally conscious marketplace.

6.3. Policy implication

The study offers several policy recommendations that support systemic CEaaS adoption in the food service sector. First, to promote sustainable sourcing, governments can introduce fiscal incentives such as tax credits or subsidies for businesses that prioritize local, regenerative, or surplus-derived ingredients. This would encourage ethical procurement and support regional food ecosystems (Chenavaz and Dimitrov, 2024). Second, the consistent importance of reusable packaging systems across datasets points to the need for regulatory interventions that facilitate waste reduction. Mandating reusable packaging use in commercial food service or introducing penalties for excessive single-use plastic can accelerate packaging circularity and reduce landfill waste (Suresh et al., 2024). Third, to support food waste mitigation, policies should promote redistribution networks and upcycling innovation. Policymakers could create digital platforms connecting food service businesses with charities and provide grants for R&D in food byproduct conversion technologies. Fourth, the importance of carbon-neutral delivery services underscores the urgency of sustainable logistics regulation. Policies could mandate emission reporting for food delivery platforms and offer incentives for adopting electric vehicles or low-emission logistics systems. Lastly, to encourage digital transformation in inventory management, policymakers can support SMEs in adopting smart technologies through funding, technical training, or tax relief. Such interventions would enhance food system efficiency while aligning national policy with consumer-driven sustainability expectations. By aligning theoretical advancement, strategic implementation, and regulatory frameworks with the prioritized CEaaS measures identified in this study, stakeholders can foster a more resilient, efficient, and sustainable food service industry.

7. Study limitation and future research direction

Despite its contributions, this study has several limitations that warrant attention. First, the survey sample, while sufficient for initial analysis, may not fully represent the diversity of global food service consumers due to potential geographic and cultural bias. Future studies should incorporate more geographically and demographically diverse populations to enhance generalizability. Second, the use of self-reported perceptions to assess CEaaS importance levels may introduce subjective bias, potentially leading to over- or underestimation of sustainability priorities. Complementing survey-based data with objective performance metrics, such as transactional records or operational sustainability outcomes, would enhance the robustness of future analyses. Third, the cross-sectional design of the study limits its ability to capture the dynamic nature of consumer attitudes toward CEaaS initiatives. Longitudinal research would be valuable to track evolving priorities over time, particularly as awareness of circular economy practices increases. Addressing these limitations can support more comprehensive, accurate, and context-sensitive insights into the adoption and effectiveness of CEaaS measures in the global food service sector.

CRediT authorship contribution statement

Tutur Wicaksono: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marhadi Marhadi:** Writing – original draft, Validation, Resources, Project administration, Investigation, Data curation. **Agustinus Fritz Wijaya:** Writing – review & editing, Software, Project administration, Methodology, Conceptualization. **Velly Anatasia:** Writing – review & editing, Software, Project administration, Conceptualization. **Krisztina Taralik:** Validation, Supervision, Project administration, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly in order to improve language and delivery fluency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be available upon request from the corresponding author.

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